

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

AN EXAMPLE OF A BAYES NET

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

Alarm Example

Pearl. Probabilistic reasoning in intelligent systems. 1988

Mr. Holmes receives a call from his neighbor Dr. Watson, who states that he hears the sound of a burglar alarm from the direction of Mr. Holmes' house. While preparing to rush home, Mr. Holmes recalls that Dr. Watson is known to be a tasteless practical joker...

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Binary variables B, A, W

$$\Pr(B = 1) = 10^{-4}$$

B	$\Pr(A = 1 B)$	A	$\Pr(W = 1 A)$
0	0.01	0	0.1
1	0.95	1	1.0

As Mr. Holmes is preparing to leave his office, he recalls that his daughter is scheduled to arrive home at any minute. If greeted by an alarm sound, she probably ($P = 0.70$) would phone him for instructions.

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New variable: D

A	$\Pr(D = 1 A)$
0	0.0
1	0.7

As he is debating whether or not to rush home, Mr. Holmes remembers reading in the instruction manual of his alarm system that the device is sensitive to earthquakes and can accidentally ($P = 0.20$) be triggered by one.

A decorative teal triangle is located in the bottom right corner of the slide.

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New variable: E

$$\Pr(E) = 0.01$$

B	E	$\Pr(A = 1 B, E)$
0	0	0.01
0	1	0.20
1	0	0.95
1	1	0.96